Alessandro **Sanvito**

Software Engineer · Systematic Trading · Machine Learning

Amsterdam, Netherlands

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Experience

Optiver
Amsterdam, Netherlands

Software Engineer, Quantitative Volatility Strategies — *Aug 2023 – Present*

- Expanded the desk into new product groups: built their baseline systematic strategies from scratch, giving researchers a validated starting point to optimize and generate new alpha.
- Owned candidate-signal selection and the data-quality and validation gates every baseline must pass before handoff to research.
- Built data pipelines over 10+ TB of instrument-level data (Spark, Polars, DuckDB), cutting core backtest runtime by 60%; migrated ML model training to AWS.

Mercedes-Benz, AI-SEE Project
Böblingen, Germany

AI Research Intern (Master's Thesis) — *May 2022 – Jul 2023*

- Researched neural scene representations for autonomous driving in Mario Bijelic's group, with Felix Heide's lab at Princeton.
- Co-authored ScatterNeRF (ICCV 2023), a NeRF that reconstructs scenes through fog, led by Andrea Ramazzina: reproduced the baselines, contributed ideas including the additive scene-plus-scattering decomposition, and wrote parts of the paper.
- First-authored HINT (IEEE IV 2024) from my thesis: complete 3D human avatars from limited viewpoints via a sagittal-symmetry prior and body-model supervision, beating state-of-the-art baselines on PSNR and SSIM.

Education

Politecnico di Milano
Milan, Italy

MSc in Computer Science and Engineering, 110/110 cum laude · double degree with KTH — *Sep 2020 – Jul 2023*

KTH Royal Institute of Technology
Stockholm, Sweden

MSc in ICT Innovation (Data Science), grade A — *Sep 2020 – May 2023*

Politecnico di Milano
Milan, Italy

BSc in Engineering of Computing Systems, 109/110 — *Sep 2017 – Jul 2020*

Publications

2024	**HINT: Learning Complete Human Neural Representations from Limited Viewpoints**, **A. Sanvito**, A. Ramazzina, S. Walz, M. Bijelic, F. Heide	*IEEE IV*
2023	**ScatterNeRF: Seeing Through Fog with Physically-Based Inverse Neural Rendering**, A. Ramazzina, M. Bijelic, S. Walz, **A. Sanvito**, D. Scheuble, F. Heide	*ICCV*
2022	**United We Stand, Divided We Fall: Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources**, P. Maldini, **A. Sanvito**, M. Surricchio	*RecSys Challenge*
2021	**Lightweight and Scalable Model for Tweet Engagements Predictions in a Resource-constrained Environment**, L. Carminati et al., incl. **A. Sanvito** · *1st among academic teams, 4th overall*	*RecSys Challenge*

Skills

Programming	Python, C++, SQL, Java, C
Machine Learning	PyTorch, NeRFs & diffusion models, computer vision, recommender systems, gradient boosting (LightGBM, CatBoost, XGBoost), scikit-learn
Data Engineering	Spark, Polars, DuckDB, Pandas, NumPy, PostgreSQL, Delta Lake, AWS
Tools	Git, Docker, Linux, CI/CD, pytest